

Roll No.:

END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: B.Tech

Semester: V

Name of the Course: Operating Systems

Course Code: TCS 502

Time: 3 Hours

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1 (20Marks)

- (a) Explain the various layers of OS and discuss the advantages and disadvantages of different OS structures in detail. CO1
- (b) What are System Calls? Explain the types of system calls used in operating systems with examples. CO1
- (c) Explain the modes of operation in vi editor. Describe at least ten commonly used commands in vi for editing and saving files effectively. CO1

Q2

- (a) Explain the different states of a process with the help of diagram. Discuss use of semaphore in dining philosopher's problem in detail. CO2
- (b) What is critical section problem? Explain various CPU scheduling criteria and discuss how they are used to evaluate the performance of different scheduling algorithms. CO2

Consider the set of 5 processes whose arrival time and burst time are given below

Process Id	Arrival time	Burst time	Priority
P1	0	4	2
P2	1	3	3
P3	2	1	4
P4	3	5	5
P5	4	2	5

- (c) If the CPU scheduling policy is priority non-preemptive, calculate the average waiting time and average turnaround time. (Higher number represents higher priority)

Q3

- Consider the following reference string: 1, 2, 3, 4, 2, 1, 6, 5, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6 calculate page hit and page miss for (No of page frames=3): CO4
- (a) (i) LRU replacement CO3
- (ii) FIFO replacement CO3
- (iii) Optimal replacement

- (b) What is Deadlock? Explain four necessary conditions for dead lock to occur with suitable example. Describe the different methods for prevention and avoidance of deadlock.

Consider the following snapshot of a system with five processes and three resources A, B, C. Using Banker's Algorithm, find the safe sequence of processes.

Process	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	1	1	2	4	3	3	2	1	0
P1	2	1	2	3	2	2			
P2	4	0	1	9	0	2			
P3	0	2	2	7	5	3			
P4	1	1	2	1	1	2			

- Q4** CO5
- (a) Describe different directory structures used in operating systems. Compare single-level, two-level, and tree-structured directories and discuss their advantages and disadvantages. CO5
- (b) Explain contiguous, linked, and indexed allocation techniques with neat diagrams and examples. CO4
- (c) Apply FCFS, SSTF, SCAN, CSCAN, LOOK and CLOOK disk scheduling algorithms on the given disk queue of requests: 82, 170, 43, 140, 24, 16, 190. Find the total seek for each case. Assume that the disk has 200 cylinders ranging from 0 to 199 and the current position of head is at cylinder 50. CO4
- Q5** CO5
- (a) Define shell functions. Explain the process of creating and executing shell scripts in Linux. CO5
- (b) Write a shell script that performs string concatenation, length calculation, and substring extraction. CO6
- (c) Explain how Linux differs from traditional monolithic kernel systems. How the Linux scheduler handles multitasking efficiently? Explain. CO6